Global Mortality Disparities: Socioeconomic and Health Patterns (2000–2019)

1. Dataset Overview

- **Source**: Kaggle – *Global Causes of Death Dataset*
- **Time Period**: 2000–2019 (Years prior to 2000 were excluded due to data inconsistencies)
- **Scope**: Annual country-level data on causes of death, enhanced with socioeconomic indicators
- **Key Variables**:
  - *Country*, *Year*, *Cause of Death*, *Death Counts*, *Continent*
  - Added metrics: *GDP per capita*, *Life Expectancy*, *Health Spending*, etc.
  - Grouping by: G7, BRICS, High-, Mid-, and Low-Income Countries
  - Calculated standardized mortality rates (per 100,000 population)
  - Computed deviation from global average deaths to reveal disparities

2. Visualization Techniques and Rationale

Each visualization was chosen to align with specific data characteristics and analytical goals:

Fourfold Plot

- Reveals the **association between disease type (chronic vs. infectious)** and **development status**
- Uses odds ratios; arc curvature and circle size represent association strength and death count

Choropleth Maps

- Visualize **geographical disparities** in chronic and infectious mortality rates
- Employ sequential red color scales for intensity mapping

Heatmap

- Tracks **temporal trends** in mortality deviation by economic group

- Utilizes a **divergent red-blue palette** to highlight deviations from the global mean

Radar Plot

- Compares **socioeconomic indicators** (GDP, Life Expectancy, Median Age, Health Spending) for five global groups across two years (2000 vs. 2019)
- Enables clear, multidimensional analysis of development trends

3. Design Considerations

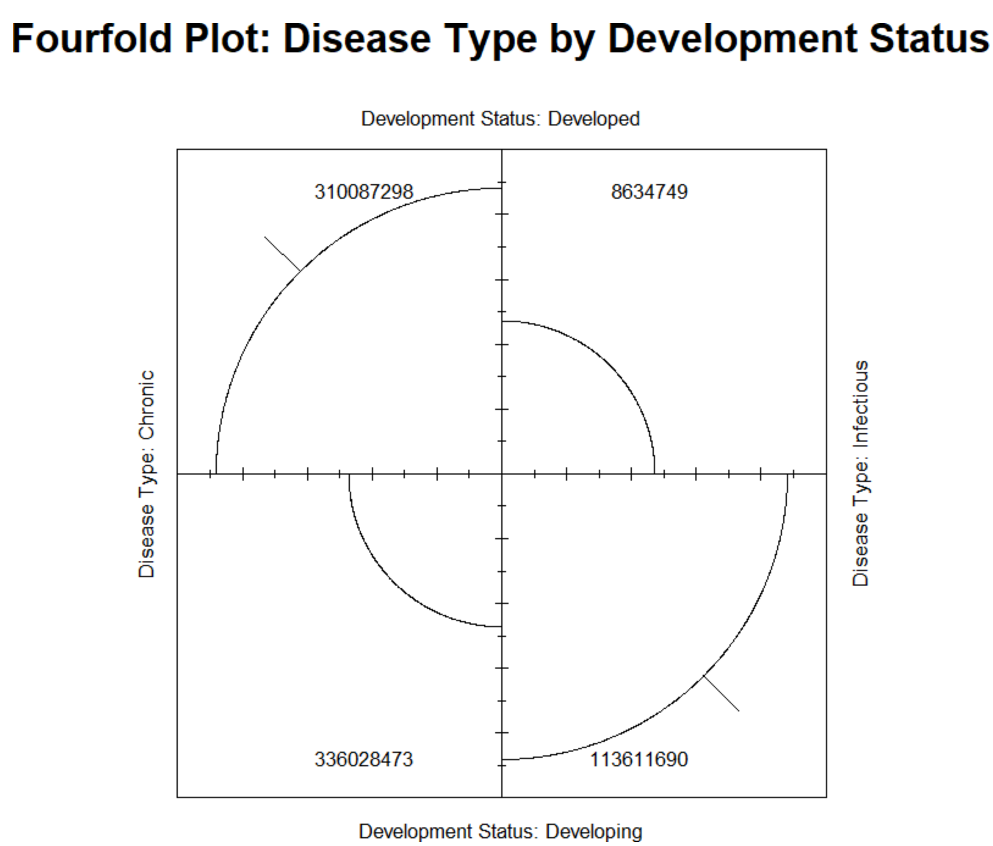
- **Color Palettes**:
  - Divergent (red-blue) for heatmaps to show deviation
  - Sequential (reds) for choropleths to emphasize magnitude
  - Consistent thematic colors across all visuals
- **Scales**:
  - Linear for raw values; per 100,000 for mortality to enable global comparison
- **Symbolization**:
  - Arc shape and radius in fourfold plots reflect statistical strength and magnitude
- **Organization**:
  - Grouping by development tier ensures structured visual comparison

Disease Burden by Development Level (Fourfold Plot)

The fourfold plot compares mortality patterns between developed and developing regions:

- **Developed Countries**:
  - Chronic disease deaths: **31 million**
  - Infectious disease deaths: **8.6 million**
- **Developing Countries**:
  - Chronic disease deaths: **33.6 million**
  - Infectious disease deaths: **11.4 million**

Key Insight:
Chronic illnesses are the primary cause of death in more affluent countries, reflecting lifestyle-related health challenges and aging populations. In contrast, infectious diseases remain a significant threat in developing regions, where limited healthcare infrastructure continues to pose challenges.

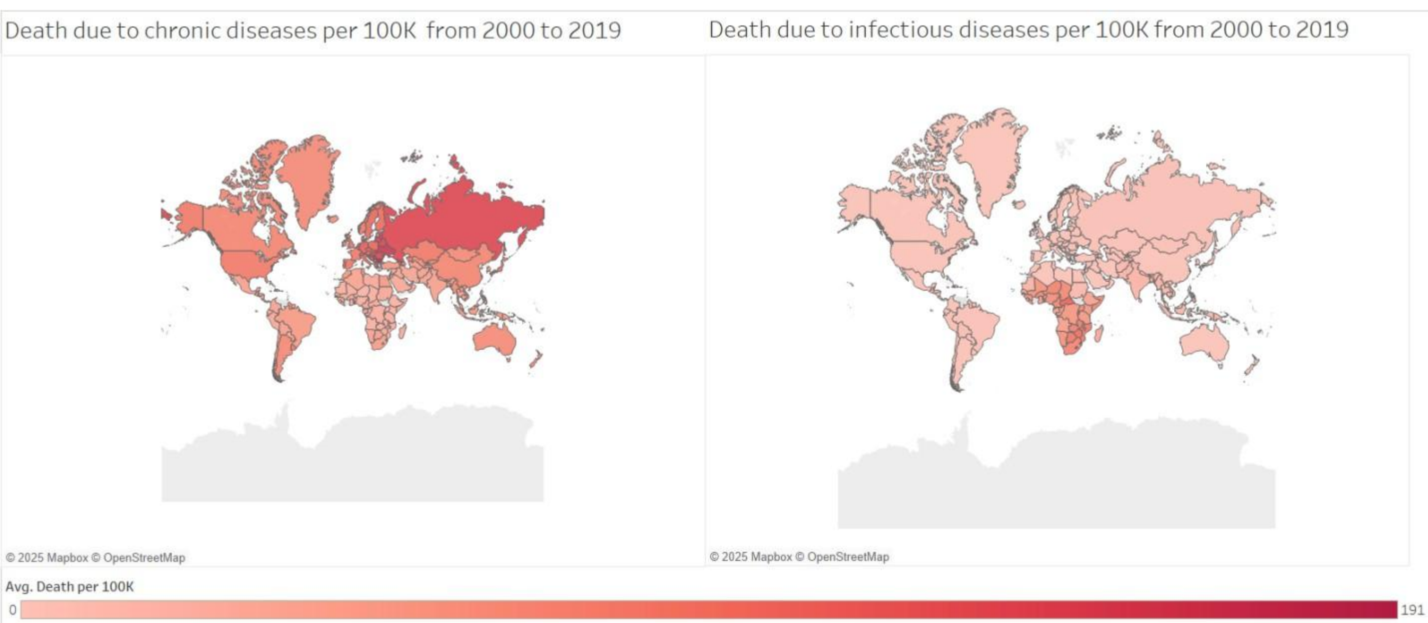


Geographic Mortality Hotspots (Choropleth Map)

Spatial analysis highlights regional concentrations of disease-related mortality:

- **Chronic Disease Mortality**:
  Highest rates are observed in **Eastern Europe and Russia**, reaching approximately **191 deaths per 100,000 people**. These areas face elevated risks due to aging demographics and non-communicable disease prevalence.
- **Infectious Disease Mortality**:
  Most severe in **Sub-Saharan Africa**, where infectious diseases continue to claim high numbers of lives due to under-resourced health systems and barriers to preventive care.

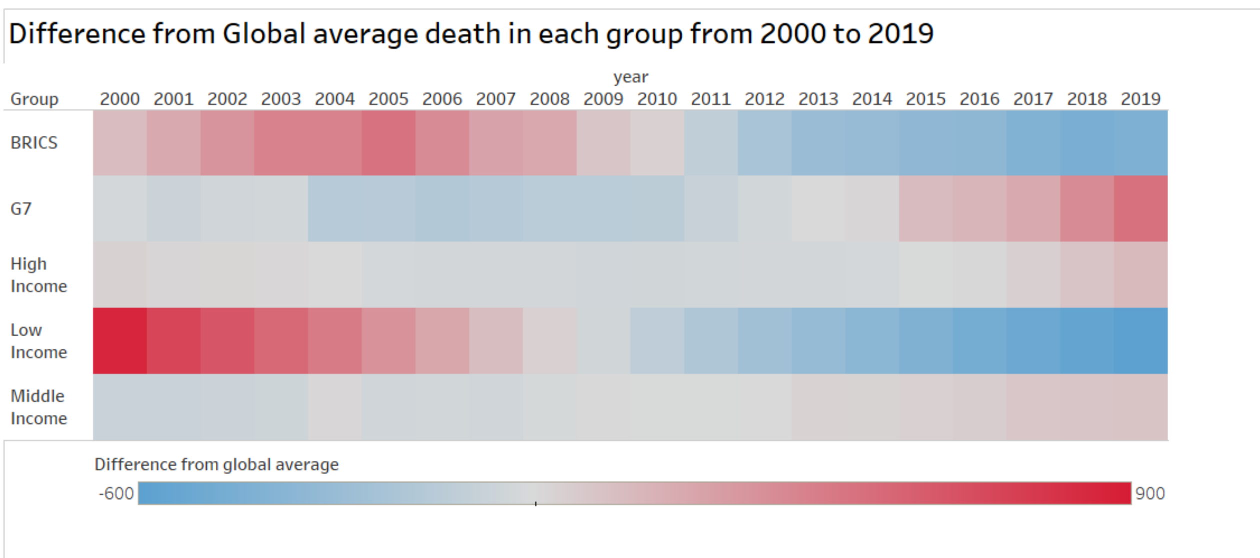
Key Insight:
While chronic diseases burden developed and aging populations, infectious diseases remain prevalent in low-resource settings. The disparity underscores the need for context-specific health interventions.



Mortality Deviation Trends Over Time (Heatmap)

- **Low-Income Countries** consistently exhibited death rates above the global average; however, these rates have shown a gradual decline over the years, indicating progress in healthcare access and quality.
- **G7 Nations** maintained lower-than-average mortality rates for most of the period but began to exceed the global average slightly after 2015—likely due to aging populations and chronic disease prevalence.
- **BRICS Countries** transitioned from having higher-than-average mortality rates in the early 2000s to falling below the global average after 2011, reflecting significant improvements in public health infrastructure.

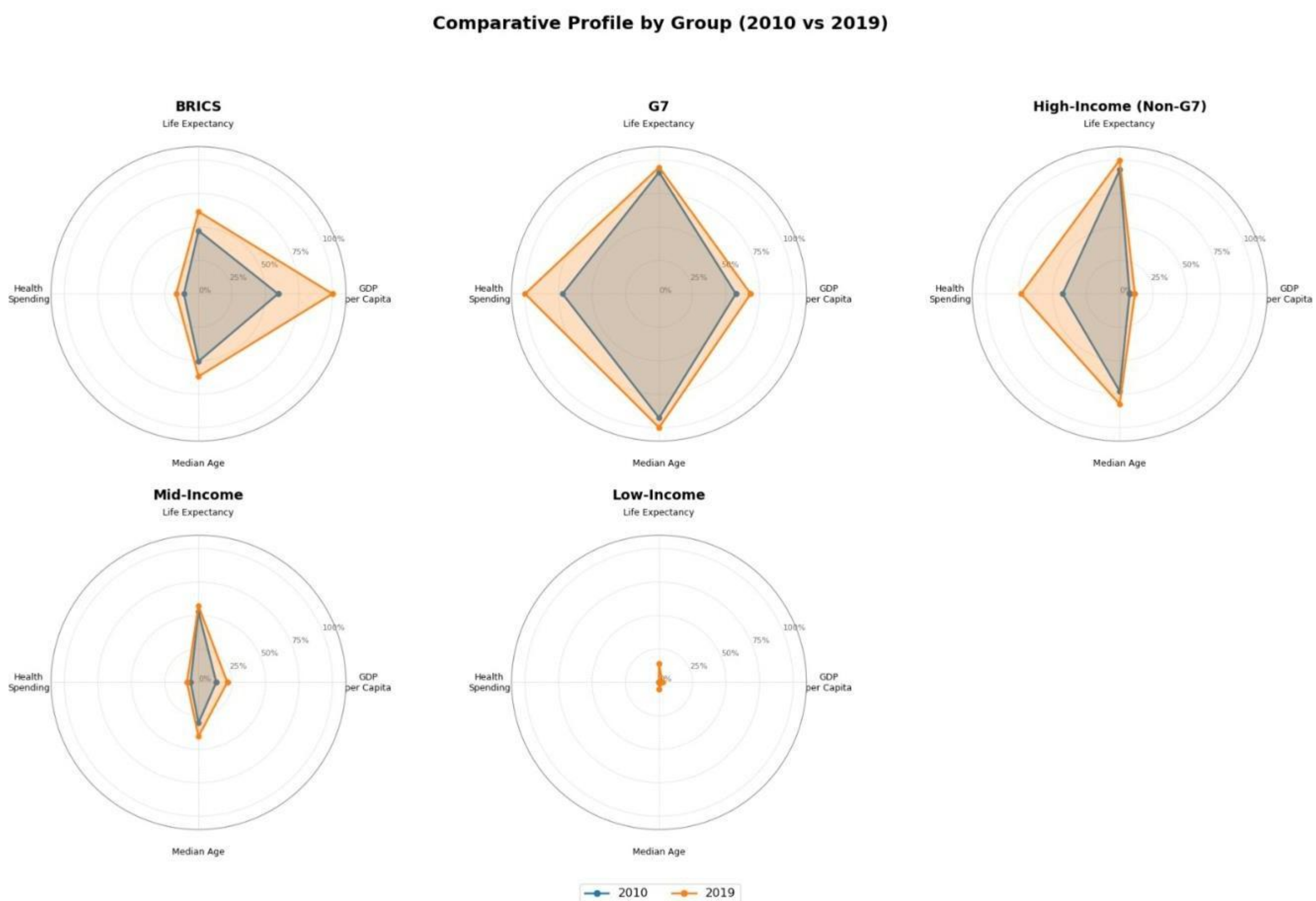
Overall Insight: The gap in mortality rates between economic groups is narrowing. Both BRICS and low-income countries have demonstrated measurable health advancements over the two decades.



Socioeconomic Development Patterns by Group (Radar Plot)

- This radar plot compares five global groups—G7, BRICS, High-Income, Mid-Income, and Low-Income—across key development indicators (life expectancy, median age, GDP per capita, and health expenditure) in **2000 and 2019**.
- **G7 countries** continue to lead across all indicators, maintaining their status as the most developed group.
- **BRICS nations** show significant improvement across all metrics, especially in GDP and health spending, signaling rapid development.
- **Low-Income countries** remain at the lower end of the spectrum but exhibit steady upward trends, suggesting slow yet consistent growth in socioeconomic conditions.

Key Insight: While disparities remain, the data reflects gradual convergence in development levels, with BRICS making the most notable strides and low-income nations showing incremental progress.



5. Conclusions and Recommendations

Key Takeaways

- Chronic diseases dominate in high-income countries; infectious diseases still impact low-income countries disproportionately.
- BRICS and low-income nations are showing **positive mortality and development trends**.
- Strong correlations observed between **socioeconomic factors** and **life expectancy**.
- Global epidemiological shift highlights the need for tailored public health policies at various stages of development.

Suggested Analytical Extensions

- **Logistic Regression**: To quantify the relationship between development status and disease type.
- **Time-Series Analysis**: To model and forecast death rate trends over time.
- **Cluster Analysis**: To identify countries with similar health and development profiles for policy targeting.

6. Future Enhancements

To expand analytical depth and engagement:

- **Interactive Dashboards**: Build with Plotly, Tableau, or Shiny for exploratory analysis
- **Predictive Modeling**: Extend regressions for future forecasting
- **Accessibility Improvements**: Use color-blind-safe palettes and responsive layouts
- **Temporal Animation**: Visualize changing disease and development trends over time